ACADEMIC CENTRAL

PRACTICE EXAMINATION

NC(V) LEVEL 3

INTRODUCTION TO ROBOTICS

PAPER 2 · PRACTICE A

MARKS: 100 **TIME: 3 hours**

NC(V) Programming and Robotics, Level 3. Paper 2 (design related). Section A: objective questions (30) · Section B: robotics and the Internet of Things, 3D printing technologies and the software pipeline, electrical power and batteries, microcontrollers, displays and wireless boards, and an Arduino design task (70).

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INSTRUCTIONS AND INFORMATION

1. Answer ALL the questions.
2. Read ALL the questions carefully.
3. Number the answers according to the numbering system used in this question paper.
4. Show ALL formulae, substitutions and calculations, and write units with all answers. Round off to TWO decimal places.
5. A non-programmable calculator may be used.
6. Write neatly and legibly.

SECTION A

QUESTION 1

1.1 Various options are given as possible answers to the following questions. Choose the answer and write only the letter (A–D) next to the question number (1.1.1–1.1.10).

1.1.1 IoT stands for …

A Integration of Technology. B Internet of Things.
C Input on Time. D Internal operating Terminal.

1.1.2 The 3D printing technology that cures liquid resin with a laser is …

A fused deposition modelling. B laminated object manufacturing.
C selective heat sintering. D stereolithography (SLA).

1.1.3 Selective laser sintering (SLS) builds parts from …

A a bed of powder. B a spool of filament.
C sheets of paper. D liquid wax only.

1.1.4 The unit used to describe how long a battery can supply a current is the …

A volt. B watt. C amp-hour. D hertz.

1.1.5 The number of AC cycles per second is measured in …

A ohms. B coulombs. C hertz. D farads.

1.1.6 A device that changes DC from a battery into AC is a/an …

A inverter. B rectifier.
C charge controller. D voltage regulator.

1.1.7 A circuit used to run a DC motor forwards and backwards from a microcontroller is a/an …

A relay only. B LDR. C ADC. D H-bridge.

1.1.8 A 16×2 LCD with an I2C interface needs only … signal wires to the Arduino.

A eight B two C sixteen D four

1.1.9 Which board has built-in Wi-Fi and Bluetooth?

A Arduino Uno B ATmega328P chip C 555 timer D ESP32

1.1.10 In Arduino, the function that runs once when the board starts is …

A loop() B setup() C main() D begin()

(10)

1.2 Indicate whether the following statements are TRUE or FALSE. Choose the answer and write only 'true' or 'false' next to the question number (1.2.1–1.2.10).

1.2.1 Robotics is an interdisciplinary field that combines mechanical, electrical and computer engineering.

1.2.2 Service robots are only used in factories.

1.2.3 A gateway connects IoT devices to the internet or the cloud.

1.2.4 Batteries connected in series add their voltages.

1.2.5 Batteries connected in parallel add their voltages.

1.2.6 An STL file describes the surface of a model as many small triangles.

1.2.7 Slicing is done after the object has been printed.

1.2.8 The neutral wire and the earth (ground) wire do exactly the same job.

1.2.9 A seven-segment display can show the digits 0 to 9.

1.2.10 analogRead() on an Arduino Uno returns a value from 0 to 1 023.

(10)

1.3 Choose a term from COLUMN B that matches a description in COLUMN A. Write only the letter (A–J) next to the question number (1.3.1–1.3.10).

	COLUMN A		**COLUMN B**
1.3.1	Robot that works for people outside factories, e.g. cleaning	A	wireless sensor network
1.3.2	Sensing the environment	B	voltage regulator
1.3.3	Deciding what to do with sensor information	C	RMS value
1.3.4	Keeps the output voltage steady	D	support structures
1.3.5	Two resistors that produce a smaller voltage	E	voltage divider
1.3.6	Prevents a solar panel from overcharging a battery	F	cognition
1.3.7	Effective (heating) value of an AC voltage	G	perception
1.3.8	Temporary material that holds up overhangs while printing	H	charge controller
1.3.9	Network of small wireless sensor nodes	I	service robot
1.3.10	analogWrite() output that controls motor speed	J	PWM

(10)

TOTAL SECTION A: [30]

SECTION B

QUESTION 2: ROBOTICS AND THE INTERNET OF THINGS

2.1 Explain the difference between a robot used for general manufacturing and a mobile robot, and give ONE example of a field robot and ONE example of a service robot. (4)

2.2 Explain how perception, cognition and actuation work together in a robot vacuum cleaner. (3)

2.3 Name and explain FOUR elements of an IoT ecosystem. (8)

2.4 Give TWO ways in which the IoT has changed the way people live or work in South Africa. (2)

[17]

QUESTION 3: 3D PRINTING

3.1 Name the 3D printing technology (FDM, SLA, SLS or DMLS) best suited to EACH part and give a reason: (a) a cheap plastic bracket for a school robot; (b) a highly detailed resin model of a robot hand; (c) a strong metal part for an aeroplane engine. (6)

3.2 Explain the steps of the 3D printing software pipeline from the 3D model to the machine instructions. (6)

3.3 Name TWO challenges or limitations of 3D printing and explain how ONE of them can be reduced. (3)

[15]

QUESTION 4: ELECTRICAL POWER

4.1 A mobile robot is powered by four 3,7 V lithium-ion cells rated at 2 500 mAh each.

4.1.1 Calculate the voltage and capacity (Ah) if all four cells are connected in series. (3)

4.1.2 Calculate the voltage and capacity (Ah) if they are connected as two parallel pairs of two cells in series (2S2P). (3)

4.1.3 The robot draws an average current of 1,25 A. Calculate how long the 2S2P battery pack will last. (2)

4.2 Differentiate between direct current and alternating current, and between single-phase and three-phase AC. (4)

4.3 Three resistors of 10 Ω, 20 Ω and 30 Ω are connected in parallel across a 12 V supply. Calculate the total resistance and the total current. (5)

[17]

QUESTION 5: COMPONENTS AND PROGRAMMING

5.1 Explain why a normal desktop computer is not suitable as the "brain" of a small mobile robot, and give FOUR criteria for choosing a microcontroller. (5)

5.2 Study the Arduino sketch below and answer the questions.

```
const int sensorPin = A0;   // light sensor (LDR voltage divider)
const int ledPin = 9;

void setup() {
  pinMode(ledPin, OUTPUT);
  Serial.begin(9600);
}

void loop() {
  int light = analogRead(sensorPin);
  Serial.println(light);
  if (light < 300) {
    digitalWrite(ledPin, HIGH);
  } else {
    digitalWrite(ledPin, LOW);
  }
  delay(200);
}
```

5.2.1 What does the robot do when it gets dark? (2)

5.2.2 What is the purpose of Serial.println(light)? (1)

5.2.3 Change the program so that the LED gets brighter as it gets darker. Write only the new loop() function. (4)

5.3 Design a "smart parking sensor" for a robot: an ultrasonic sensor measures the distance to a wall, a 16×2 I2C LCD shows the distance in cm, and a buzzer sounds when the wall is closer than 20 cm. List the components and describe, in steps, how the program must work. (6)

5.4 Compare a seven-segment display and a 16×2 LCD for showing robot information. (3)

[21]

TOTAL SECTION B: [70]

TOTAL: 100

END OF QUESTION PAPER